Consistency Models

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Abstract

Diffusion models generate very good images, but they need many network evaluations to make a single image, which makes them slow. Consistency models [1] are trained to jump from a noisy image straight to the clean one, so they can generate an image with one evaluation. In this project we reproduce them on a small scale with MNIST. We train two "teacher" models (a diffusion model and a Rectified Flow model), distill each one into a consistency model, and also train consistency models from scratch, without a teacher. We compare everything in quality, diversity and speed, using a distance computed with our own digit classifier. With one evaluation, the distilled models are 50 to 95 times faster than the teachers with 50 steps, but their images are clearly worse: their distance is about 3 and 5 times larger than that of their teacher (2.2 and 3.4 times with four steps). The Rectified Flow teacher is much better than the diffusion one, but both distilled models end up with about the same quality, so the teacher does not seem to be what limits them. Our first training from scratch worked badly, but a second run closer to the original paper, with the training images of distillation, is about as good as the distillation of the diffusion teacher. An improved recipe, with the same images, is clearly better with two and four steps.

1 Introduction

Diffusion models [2, 3] are one of the best ways to generate images today. They learn to remove noise: to create an image, we start from pure noise and clean it little by little, running the neural network at every step. This is slow. The original model [2] uses 1000 steps per image, and later methods based on the *probability flow ODE* [3] reduce this to tens of steps, which is still a lot.

Consistency Models [1] were proposed to solve this. A consistency model takes a noisy image from any point of the denoising path and jumps directly to the clean image at the end of the path, so a single network evaluation is enough. If there is more time, we can use several steps (adding some noise back and cleaning again) to get better images. They can be trained in two ways: *consistency distillation* (CD), where a pre-trained diffusion model acts as a *teacher*, and *consistency training* (CT), where no teacher is used.

In this project we reproduce the method on a small scale with MNIST [4], a dataset light enough to train several models on a CPU, and we measure what is gained and what is lost when fewer steps are used. Besides the initial proposal, we also trained a Rectified Flow [5] teacher and distilled it too, to see how the choice of teacher affects the result.

2 Objectives

The main objectives set in the project proposal are:

1. Train a diffusion model to serve as the teacher of the consistency model.
2. Distill the knowledge learned by the teacher into a faster consistency model.
3. Compare the consistency model with the diffusion model in terms of sampling speed and sample quality.
4. Analyze the results: the speed-up, the quality and diversity of the samples with more or fewer steps, and the limitations we found.

Once these objectives were met, the proposal considered three extensions, in this order: consistency training and its comparison with the previous models; adapting the architecture to color images with CIFAR-10 [6]; and a class-conditional consistency model. Their status is given in Section 6.

3 Related work and state of the art

Diffusion models. Diffusion models [7, 2, 8] were put in a common framework by Song et al. [3] using stochastic differential equations; each one has a deterministic version, the probability flow ODE. Karras et al. [9] studied the design choices (noise schedule, network inputs, ODE solver), and their conclusions are the basis of consistency models and of our code.

Fast sampling. Some methods speed up sampling without extra training, using better ODE solvers such as DDIM [10], the Heun method of EDM [9] or DPM-Solver [11], which gets good quality in around 10 steps. Progressive distillation [12] halves the number of steps in each round, down to 4, but needs several training rounds. Flow matching [13] and Rectified Flow [5] learn paths that are almost straight, which can be followed in fewer steps, and *reflow* straightens them further.

Consistency models. Consistency models [1] learn to jump from any point of the path to its origin, allow one-step generation, and are trained in a single stage, by distillation (CD) or from scratch (CT). Song and Dhariwal [14] improved CT a lot with several changes. Later works apply the idea to text-to-image models [15], to jumps between any two points of the path [16] and to continuous time with 1.5 B parameters [17]. All of them use large GPU budgets; ours is a small reproduction on a CPU that also compares a diffusion and a Rectified Flow teacher.

4 Methodology

We train two teachers on MNIST, a diffusion model of the Variance Exploding type (VE) and a Rectified Flow (RF), and distill each of them into a consistency model. We also train consistency models from scratch, without a teacher: the original method (CT) and an improved recipe (iCT), each in two versions, v1 and v2 (Table 1 summarizes them). All models use the same network, so differences between them come from how they are trained and not from their size. The code, written in PyTorch, and a tutorial notebook are available at `https://github.com/ibon44/Consistency_model_genai`.

4.1 Data and architecture

We use the 60 000 MNIST training images (28×28, grayscale, values scaled to $[-1, 1]$, no data augmentation). The 10 000 test images are only used for evaluation (Section 4.6).

All models use the same small U-Net [18] with 1 115 425 parameters (details in Appendix A.1). Only the way its output is used changes: the score network divides it by the noise level $\sigma(t)$, the velocity network uses it as it is, and the consistency models wrap it as explained in Section 4.3.

4.2 Teacher models

In both teachers the clean image is at time $t = 0$ and pure noise at $t = T = 1$, and the training times are drawn from $[\varepsilon, 1]$ with $\varepsilon = 10^{-5}$. Using the same convention lets us train and sample the consistency models with the same code for both teachers.

VE diffusion. The forward process only adds Gaussian noise, $x_t = x_0 + \sigma(t)\, z$ with $z \sim \mathcal{N}(0, I)$ and

$$\sigma(t) = \sqrt{\frac{\sigma^{2t} - 1}{2 \ln \sigma}}, \qquad \sigma = 25, \tag{1}$$

which gives a noise standard deviation of about 9.85 at $t = 1$. The network s_θ (the score) is trained to predict the noise that was added [3], minimizing $\mathbb{E}\, \|\sigma(t)\, s_\theta(x_t, t) + z\|^2$ with t uniform in $[\varepsilon, 1]$. The teacher is trained for 60 epochs with Adam, batch size 64 and a learning rate of 10^{-3} with cosine decay to zero, and we keep a smoothed copy of the weights (an exponential moving average, EMA, with rate 0.999). An earlier teacher, trained for 40 epochs with a constant learning rate and without EMA, was clearly worse; we only use it in an ablation (Appendix A.11). To generate images we solve the probability flow ODE

$$\frac{dx}{dt} = -\frac{1}{2}\, g(t)^2\, s_\theta(x, t), \qquad g(t) = \sigma^t, \tag{2}$$

from $t = 1$ to $t = \varepsilon$ with Heun's method on the time grid of Karras et al. [9] ($\rho = 7$), spaced in σ and not in t. Each Heun step uses two network evaluations.

Rectified Flow. Rectified Flow [5] joins each image to a noise sample $x_1 \sim \mathcal{N}(0, I)$ with a straight line, $x_t = (1 - t)\, x_0 + t\, x_1$, and trains a velocity network v_θ to predict the (constant) velocity of that line:

$$\mathcal{L}_{\text{RF}} = \mathbb{E}\, \big\| v_\theta(x_t, t) - (x_1 - x_0) \big\|^2, \qquad t \sim \mathcal{U}[\varepsilon, 1]. \tag{3}$$

It is trained for 40 epochs with Adam, batch size 64 and a constant learning rate of 10^{-3}, also with an EMA of the weights (rate 0.999). To generate images we integrate $dx/dt = v_\theta(x, t)$ from $t = 1$ to $t = 0$ with equal Euler steps, one network evaluation each. Heun steps and the cosine decay used for the VE teacher did not improve the samples. The two teachers were tuned separately and with different budgets (60 epochs for VE, 40 for RF), which must be kept in mind when comparing them.

The learned paths are not exactly straight, because the lines of different image–noise pairs cross and the network only learns their average. To straighten them we do one round of *reflow* [5]: we map 60 000 noise samples to images with the first model (1-RF, EMA weights, 100 Euler steps) and retrain on these fixed pairs, whose lines cross much less. The resulting model, the 2-RF, starts from the 1-RF weights and is trained for 10 epochs with a learning rate of $2 \cdot 10^{-4}$.

4.3 Consistency models

Parameterization. A consistency model $f_\theta(x_t, t)$ must give the same clean image for every point of a path, and it must return the input unchanged at $t = \varepsilon$, that is, $f_\theta(x, \varepsilon) = x$. As in the original paper, we guarantee this by construction:

$$f_\theta(x, t) = c_{\text{skip}}(t)\, x + c_{\text{out}}(t)\, F_\theta\big(c_{\text{in}}(t)\, x,\ t\big), \tag{4}$$

where F_θ is the U-Net of Section 4.1. At $t = \varepsilon$, $c_{\text{skip}} = 1$ and $c_{\text{out}} = 0$, so the output is exactly x (coefficients in Appendix A.2).

Consistency distillation. We follow Algorithm 2 of Song et al. [1]. We split $[\varepsilon, T]$ into $N = 150$ times with the Karras spacing (in σ for VE and directly in t for RF). At each iteration, for each image x_0 of the batch, we pick two neighbouring times $t_n < t_{n+1}$, add noise to the image up to $x_{t_{n+1}}$ with the forward process of the teacher, and use the teacher to take one ODE step back to $\hat{x}_{t_n}$ (Heun for VE, Euler for RF). Both points are, approximately, on the same path, so the loss asks the model to give the same output at both:

$$\mathcal{L}_{\text{CD}} = d\big(f_\theta(x_{t_{n+1}}, t_{n+1}),\ f_{\theta^-}(\hat{x}_{t_n}, t_n)\big). \tag{5}$$

The target network f_{θ^-} is an EMA of f_θ (rate $\mu = 0.95$) and receives no gradients. As the distance d we use the pseudo-Huber distance $d(a, b) = \sqrt{\|a - b\|^2 + c^2} - c$ with $c = 0.00054\sqrt{784} \approx 0.015$ [14]. We train for 40 000 iterations with Adam, a constant learning rate of $4 \cdot 10^{-4}$ and batch size 64. We also keep a slower EMA of the weights (rate 0.9999), which is the one saved and used for sampling.

The value of N is our main change with respect to the paper, which uses $N = 18$. During training the network only sees these N times. With $N = 18$ our small U-Net did not work at the times in

between: one-step samples (always at $t = T$) looked fine, but multistep sampling, which adds noise at other times, gave unrecognizable images. With $N = 150$ the problem disappeared.

We distill three models: CM-VE from the VE teacher, CM-RF from the EMA weights of the 1-RF, and CM-RF (reflow) from the EMA weights of the 2-RF (recipes in Appendix A.3). A first CM-VE, distilled from the earlier VE teacher without EMA, is reported in Appendix A.11.

Consistency training. CT (Algorithm 3 of the original paper) trains the model without a teacher, and we only implemented it for the VE process. Instead of the teacher's ODE step, both points of the pair are built with the same noise: $x_{t_{n+1}} = x_0 + \sigma(t_{n+1})\, z$ and $x_{t_n} = x_0 + \sigma(t_n)\, z$, which lie (in expectation) on the same path. The loss and the distance are those of CD, but N and μ follow the schedule of the paper: both grow during training, up to $N = 151$ (formulas in Appendix A.4). We use Adam, batch size 64 and a learning rate of $4 \cdot 10^{-4}$. CT v1 follows the recipe of the paper with a short budget: 20 000 iterations, cosine decay, $\mu_0 = 0.95$, and sampling with the target network. CT v2 was designed after seeing that CT v1 was far behind CD. It moves closer to the paper's CIFAR-10 setup and to the budget of CD: 40 000 iterations, constant learning rate, $\mu_0 = 0.9$, and a separate EMA of the weights for sampling (rate 0.9995).

Improved consistency training. Because CT v1 was far behind CD, we also trained a model with the main ideas of Song and Dhariwal [14] (formulas in Appendix A.4): no EMA target (the target is the same network, without gradients), a number of times N that doubles in equal stages, noise levels drawn from a lognormal distribution that favours intermediate values, and a loss weighted by $1/(\sigma_{n+1} - \sigma_n)$. Our main run, iCT v2, has the budget of CD and CT v2 (2.6 million training images): 40 000 iterations, constant learning rate, N from 11 to 161 and a sampling EMA of rate 0.9995. An earlier run, iCT v1, used two phases (Appendix A.3): 20 000 iterations with N up to 161 (stage 1), and 32 289 more with batch size 128 and N up to 641, for about 5.4 million images. Table 1 lists the four teacher-free models next to CD. The v2 runs were designed after seeing the results of the v1 runs and change several settings at once, so we cannot say which change explains the improvement.

Table 1: Consistency models trained without a teacher, in order of development, and CD for reference (all with batch size 64 unless stated).

Model	Recipe	Iterations	Images
CT v1	original CT: cosine lr, $\mu_0 = 0.95$, target-network sampling	20 000	1.3 M
CT v2	constant lr, $\mu_0 = 0.9$, sampling EMA 0.9995	40 000	2.6 M
iCT v1	improved loss; N up to 161, then to 641 (batch 128)	20 000 + 32 289	5.4 M
iCT v2	improved loss; N up to 161, constant lr, EMA 0.9995	40 000	2.6 M
CD (for reference)	with the VE teacher, $N = 150$, $\mu = 0.95$	40 000	2.6 M

4.4 Sampling from consistency models

To generate an image in one step we draw $x_T \sim \mathcal{N}(0, \sigma(T)^2 I)$, with $\sigma(T) \approx 9.85$ for VE and $\sigma(T) = 1$ for RF, and compute $f_\theta(x_T, T)$: one network evaluation. With more steps we use Algorithm 1 of the original paper: for some times $\tau_1 > \cdots > \tau_{n-1}$ we add noise back to the current prediction (clamped to $[-1, 1]$) and clean it again in one jump, so n steps cost n evaluations. The noise is added with the forward process of the teacher, so that x_τ looks like the inputs seen in training (Appendix A.2). The paper chooses the times τ_i with a search that minimizes FID; we did not do this. For VE we use the interior points of a Karras grid in σ with $n + 1$ points ($\tau = 0.16$ for two steps and $\tau = 0.60, 0.16, 0.004$ for four). For RF this rule adds almost no noise (it gives $\tau = 0.027$ for two steps), so we space the times uniformly from $\tau_{\max} = 0.5$ down to ε ($\tau = 0.5, 0.33, 0.17$ for four steps). We chose $\tau_{\max}$ by looking at samples for values between 0.1 and 0.9 (Appendix A.6).

4.5 Differences from the original paper and limitations

Besides the much smaller scale (1.1 M parameters, 40 000 iterations), our work differs from Song et al. [1] in these points: the diffusion teacher is the VE SDE [3] and not an EDM model, and the Rectified Flow teacher is an addition; we use $N = 150$ instead of 18, the pseudo-Huber distance

instead of LPIPS or ℓ_2, and re-noising times that are not optimized for FID; the EDM coefficients, designed for $x_t = x_0 + \sigma(t)\, z$, are reused for RF (where the image is also scaled by $1 - t$) so that both teachers share one consistency model, which is reasonable but not optimal; and CT and iCT are only implemented for VE (iCT uses four of the ideas of [14], not dropout or the Fourier embedding scale, and reaches $N = 161$, or 641 in the earlier run, instead of 1280). Instead of FID we use a Fréchet distance on the features of our own classifier (Section 4.6).

4.6 Evaluation protocol

Reference classifier. Inception-based metrics such as FID are designed for natural images, so we trained a small CNN on MNIST (two 3×3 convolutional blocks with 32 and 64 channels, a dense layer of 128 units and 10 outputs; 421 642 parameters; 10 epochs of Adam with learning rate 10^{-3} and batch size 128). It reaches 99.00% accuracy on the test set.

Quality and diversity. Our main quality metric is FD-MNIST, a Fréchet distance computed like FID [19] but using the features of our classifier (its 128-dimensional penultimate layer) instead of Inception, and 10 000 training images as the real reference. Lower is better. It depends on the classifier we chose, so it is not comparable with published FIDs, and we use it together with a visual inspection of the samples. As a second metric we report the fraction of *recognizable* samples (the classifier gives some digit a probability of at least 0.9). The classifier is overconfident with images it has never seen and gives such a probability to 61% of pure-noise images, so this metric must be read with care. Diversity is the entropy of the distribution of predicted digits, with a maximum of $\ln 10 \approx 2.30$ when all digits are equally frequent. Since FD depends on the number of samples, every configuration is evaluated on the same $N_{\text{eval}} = 2000$ samples, generated from the same initial noise within each family (VE or RF). The references are 2000 real test images (the best possible values) and 2000 images of Gaussian noise clamped to $[-1, 1]$.

Timing. We measure the wall-clock time to generate $N \in \{1, 10, 100, 500\}$ images (mean of 3 runs after a warm-up) and report the time per image at $N = 500$, always on the same CPU. The time only depends on the network and on the number of evaluations, so we measured it once per family and number of steps and shared it between models of the same kind. The speed-up is relative to the 50-step teacher of the same family. We also report the number of network evaluations (NFE) per image, which does not depend on the hardware.

5 Results and discussion

Table 2 summarizes quality, diversity and speed; the complete table is in Appendix A.7. Each model was trained only once, so we cannot measure how much the results change between runs. We can measure the noise coming from sampling: evaluating the same models on five other noise draws changes FD by up to 13 (Appendix A.8), so differences smaller than about 25 should not be interpreted.

5.1 Quality versus number of steps

Real test images get an FD-MNIST of 45.1, the best value we can reach with 2000 samples, and pure noise gets 2947.5. The teachers need about 10 steps to make good images (Figure 1). The 1-RF teacher gets 152 with 10 evaluations and 78.5 with 50. The VE teacher is unusable with 5 Heun steps (902) and stays at 120–126 from 10 steps on. With one step the 1-RF teacher only outputs the average of the data, a blurry blob (3791), while the straighter 2-RF gets 534.

The distilled models make digits with one evaluation (418 for CM-VE and 417 for CM-RF) and every extra step improves them: 346 and 279 with two and four steps for CM-VE, and 342 and 264 for CM-RF. For a lower cost they are much better than their teachers with few steps (CM-VE with 4 evaluations gets 279, while the VE teacher gets 902 with 10), but they do not reach their quality. Compared with its teacher (50 steps), CM-VE is 3.3 times worse with one step and 2.2 times with four steps, and CM-RF is 5.3 and 3.4 times worse. The two distilled models end up with almost the same FD even though the RF teacher is much better (78.5 against 126; see the Discussion). The

Table 2: Summary of the comparison ($N_{\text{eval}} = 2000$ samples per row). FD-MNIST: lower is better; recognizable and entropy: higher is better. Time per image at $N = 500$ on CPU; speed-up with respect to the 50-step teacher of the same family. CM-VE (CD) was distilled from the VE teacher, CM-RF from the 1-RF and CM-RF (reflow) from the 2-RF; CT and iCT use no teacher.

Model	Steps	NFE	FD-MNIST	Recogn. (%)	Entropy (%)	ms/img	Speed-up
Real MNIST (test)	–	–	45.1	97.6	100	–	–
Pure noise	–	–	2947.5	61.0	23	–	–
VE teacher	10	20	120.4	79.9	98	13.26	×5.0
VE teacher	50	100	125.7	80.8	98	66.43	×1.0
CM-VE (CD)	1	1	418.3	65.4	97	0.70	×95.4
CM-VE (CD)	4	4	279.4	73.5	98	2.68	×24.8
CM-VE (CT v1)	1	1	1922.4	38.5	50	0.70	×95.4
CM-VE (CT v1)	4	4	910.6	50.9	87	2.68	×24.8
CM-VE (CT v2)	1	1	453.5	65.5	98	0.70	×95.4
CM-VE (CT v2)	4	4	267.7	74.0	99	2.68	×24.8
CM-VE (iCT v1)	4	4	174.4	78.1	98	2.68	×24.8
CM-VE (iCT v2)	1	1	417.6	65.4	98	0.70	×95.4
CM-VE (iCT v2)	4	4	163.3	76.8	99	2.68	×24.8
RF teacher (1-RF)	1	1	3791.3	0.0	14	0.66	×50.1
RF teacher (1-RF)	10	10	151.9	78.4	99	6.61	×5.0
RF teacher (1-RF)	50	50	78.5	83.0	100	33.26	×1.0
RF teacher (2-RF)	1	1	533.6	61.4	98	0.66	×50.1
RF teacher (2-RF)	50	50	160.0	76.2	99	33.26	×1.0
CM-RF (CD)	1	1	417.0	66.4	98	0.67	×49.6
CM-RF (CD)	4	4	263.6	74.3	99	2.68	×12.4
CM-RF (reflow)	1	1	394.3	66.6	99	0.67	×49.6
CM-RF (reflow)	4	4	536.8	70.1	99	2.68	×12.4

recognizable fraction of the one-step models (65–66%) is close to the 61% of pure noise, so this metric says little about them and we rely on FD and on the images.

Reflow barely changes the one-step model (FD 394 against 417, at the limit of the sampling noise), but it makes the multistep ones *worse* (521 and 537 against 342 and 264). Visually, with two and four steps the reflow samples have fainter, less saturated strokes (Figure 2). The 2-RF teacher is worse than the 1-RF with many steps (160 against 78.5 with 50), but this does not explain the result: CM-RF (reflow) already beats its teacher with one step (394 against 534), and then gets worse with more steps while its teacher gets better. We did not investigate the cause; one possibility is that $\tau_{\max}$ was chosen for CM-RF and not for the reflow model. Distilling from the raw weights of the 1-RF instead of the EMA ones gives similar samples but a slightly worse FD (Appendix A.10).

Consistency training without a teacher. CT v1 is far from distillation: FD 1922 with one step (noisy blobs) and 911 with four (Table 4). CT v2 gets 454, 353 and 268 with one, two and four steps. With the same 2.6 million training images, it is slightly worse than CM-VE (CD) with one step (454 against 418; 38 more in the five-seed means) and equal with two and four (353 and 268 against 346 and 279). iCT v2, with the same images, gets 418, 270 and 163, and is clearly better than CM-VE (CD) with two and four steps; with four evaluations that is 1.3 times the FD of the VE teacher with 50 steps (126). iCT v1, with 5.4 million images, gets 424, 270 and 174 (692, 427 and 248 after its first stage), less than 13 FD from iCT v2 over five noise seeds, so doubling the images did not help, although other settings also differ. CT v1 saw only 1.3 million images, half of the other runs.

5.2 Sampling speed

The time per image is dominated by the network evaluations, so the speed-up of each model is almost exactly the ratio of evaluations (Table 2). With one step, CM-VE and CM-RF take 0.70 and 0.67 ms per image, which is $\times 95$ faster than the 50-step VE teacher (66.4 ms, 100 evaluations) and $\times 50$ faster than the 50-step RF teacher (33.3 ms). The total time grows roughly linearly with the number of

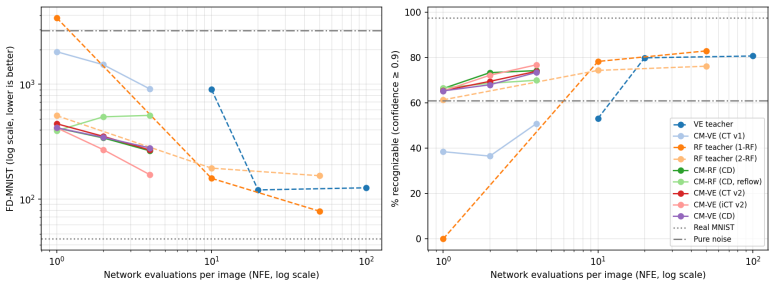


Figure 1: Quality of the samples against the number of network evaluations (NFE): FD-MNIST (left, lower is better) and percentage of recognizable samples (right). Dashed lines are teachers and solid lines consistency models; gray lines are the real-MNIST and pure-noise references.

images N (Figure 3, Appendix A.5), so the speed-ups of the tables ($N = 500$) also hold for batches of ten or more images; for a single image fixed overheads make them smaller ($\times 75$ for VE and $\times 43$ for RF with one step). These numbers must be read together with the previous section: a 10-step teacher is already as good as a 50-step one for VE and within a factor of two for RF, so compared with it (FD 120 for VE and 152 for RF) the one-step speed-up is only $\times 19$ and $\times 10$, with an FD of 418 and 417.

5.3 Diversity

The digits predicted by the classifier (entropy column of Table 2, per-digit shares in Appendix A.7) are close to uniform for the RF teachers with 10 or more steps (99–100% of the maximum entropy) and for all CM-RF models (98–99%), so we see no collapse onto a few digits (the measure cannot detect a lack of variety within a digit). The VE family is less balanced: the VE teacher gives only 4% of ones and 16% of twos (real MNIST has 12% and 11%), and CM-VE (CD), CT v2 and iCT show a similar lack of ones (3–5%), with 97–99% entropy. The low entropy of the one-step 1-RF teacher (14%) and of the one-step CT v1 model has a trivial cause: the first outputs almost the same blurry image for every noise (a 3 in 92% of the cases) and the second noise-like images that the classifier labels as 2 or 8 (88%), as it does with pure noise.

5.4 Visual inspection

Figure 2 shows three initial noises (ten in Appendix A.9). One-step consistency models make recognizable digits with ragged edges and broken strokes, and more steps clean the strokes and the background (see the 5 of the RF family, third row). They can also damage the digit: with four steps CM-VE (CD) leaves the 4 of the second row and the 5 of the third row with thin, broken strokes, CT v2 turns the 4 into a 9-like shape, and iCT v2 turns it into a shape that is no longer a digit. The VE consistency models are visibly noisier than the RF ones; CT v1 gives noisy blobs even with four steps, CT v2 is comparable to CM-VE (CD), and the four-step iCT samples are the cleanest of the VE models, although many digits are still deformed.

5.5 Discussion

Teacher choice. CM-RF and CM-VE reach about the same quality (FD 417 and 418 with one step, 342 and 346 with two, 264 and 279 with four; the differences are within the sampling noise), although the RF teacher is much better than the VE one (FD 78.5 against 126). Compared with its own teacher, CM-VE is the closer copy (3.3 and 2.2 times its FD with one and four steps, against 5.3 and 3.4 for CM-RF). So a better teacher did not give a better distilled model, and inside the VE family a teacher whose FD went from 206 to 126 improved CM-VE by only 36 to 61 (Appendix A.11). This suggests that, above some quality, what limits our distilled models is something shared by all the runs (the small network, 2.6 million images, the distance) and not the teacher; CT v2 and iCT v2, without a teacher, reach the same one-step FD (454 and 418), although we did not test this directly. The comparison is not clean anyway: the teachers had different budgets, RF distillation uses Euler steps while VE uses Heun, and each model was trained once.

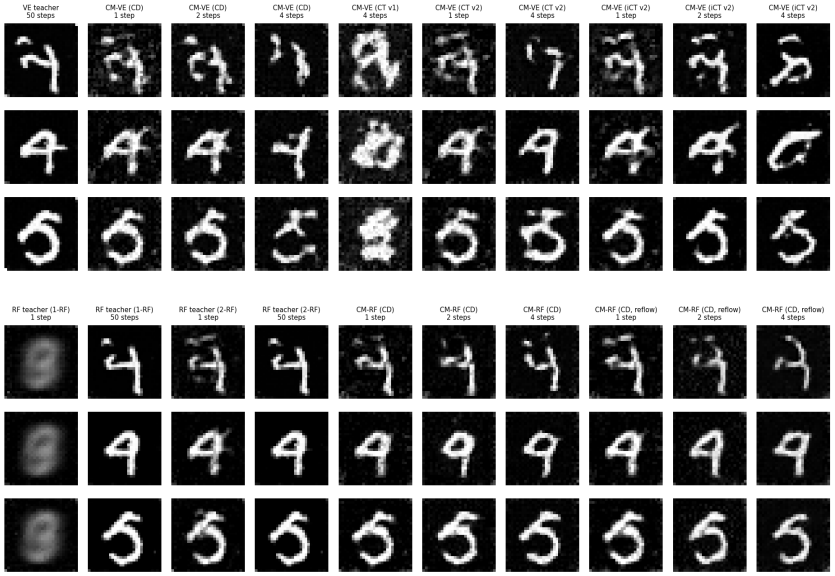


Figure 2: Samples from the VE family (top) and the RF family (bottom). Each column is a configuration and each row starts from the same initial noise.

Training without a teacher. CT v1 suggested that CT is far worse than distillation, but CT v2 shows that most of that gap came from its settings and budget: with the images of CD it is about as good as CM-VE (CD), slightly worse with one step, and iCT v2 beats CD clearly with two and four steps (270 and 163 against 346 and 279). So in our setting a teacher was not needed to match distillation, but this may not hold in general: the CD recipe was kept from its first working version, while the v2 runs were designed after seeing the v1 results.

Limitations. Besides Section 4.5: every model is a single training run; FD-MNIST uses 2000 samples and our own classifier, so it is not comparable with published FIDs; the timings come from one CPU; and the re-noising times were chosen by eye only for RF, while VE uses a fixed rule, so the multistep results of the two families are not tuned equally.

6 Conclusions and future work

We trained a VE diffusion model and a Rectified Flow model on MNIST, distilled both into consistency models and compared them in quality, diversity and speed, so the four main objectives of the proposal are met. Of the extensions, we did consistency training, in two runs of the original method and with an improved recipe; color images (CIFAR-10) and conditional generation were left out for lack of time. In addition, following a suggestion made during the course, we added the Rectified Flow teacher and a reflow round.

The main results are: (i) one-step consistency models make recognizable digits about 50 to 95 times faster than the 50-step teachers, and more steps improve them, except for CM-RF (reflow); (ii) with four steps their FD-MNIST is 2.2 (VE) and 3.4 (RF) times worse than that of their 50-step teachers, and teachers with 10 evaluations (RF) or 20 (VE) are already better; (iii) CM-RF and CM-VE reach about the same FD even though the RF teacher is much better, so the teacher does not seem to be the main limit; (iv) reflow barely changes one-step sampling and hurts multistep sampling with the re-noising times chosen for CM-RF; (v) our first CT is far behind CD, but CT v2, with the same training images, is about as good as CM-VE (FD 268 against 279 with four steps), and iCT v2 reaches 163.

Future work includes testing whether a larger network, more training images or a perceptual distance raise the limit shared by our models, deriving c_{skip} and c_{out} for the RF interpolation, choosing the re-noising times by FD, repeating the trainings with several seeds, and color images.

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A Technical appendices

A.1 U-Net architecture

The encoder has four convolutional levels with 32, 64, 128 and 256 channels (3×3 kernels, no padding, stride 1 in the first level and stride 2 in the other three), and the decoder mirrors it with transposed convolutions; the outputs of the first three encoder levels are concatenated to the inputs of the last three decoder levels. Each level is followed by group normalization (4 groups in the first level, 32 in the others) and the SiLU activation. The time t is encoded with Gaussian random Fourier features (128 fixed random frequencies with scale 30, giving 256 sine/cosine features) followed by a linear layer, and the result goes through SiLU and a dense layer per level whose output is added to the feature maps after every convolution except the last one. Of the 1 115 425 parameters, 128 (the random frequencies) are not trainable. The score network divides the output by $\sigma(t)$; the velocity network and the consistency models use it as it is.

A.2 Consistency model coefficients and re-noising

With $\sigma(t)$ the noise level of the teacher (the VE one, or $\sigma(t) = t$ for RF) and $\Delta(t) = \sigma(t) - \sigma(\varepsilon)$, the coefficients are

$$c_{\text{skip}}(t) = \frac{\sigma_{\text{d}}^2}{\Delta(t)^2 + \sigma_{\text{d}}^2}, \quad c_{\text{out}}(t) = \frac{\sigma_{\text{d}}\,\Delta(t)}{\sqrt{\sigma(t)^2 + \sigma_{\text{d}}^2}}, \quad c_{\text{in}}(t) = \frac{1}{\sqrt{\sigma(t)^2 + \sigma_{\text{d}}^2}}. \tag{6}$$

We use $\sigma_{\text{d}} = 0.5$ as in the original paper, although the standard deviation of MNIST in $[-1, 1]$ is closer to 0.62. The input scaling c_{in} comes from EDM [9]; without it the network would receive inputs with a standard deviation of up to 10 with the VE teacher. In multistep sampling, the re-noising with $z \sim \mathcal{N}(0, I)$ is

$$\text{VE: } x_\tau = x + \sqrt{\sigma(\tau)^2 - \sigma(\varepsilon)^2}\; z, \qquad \text{RF: } x_\tau = (1 - \tau)\, x + \tau\, z. \tag{7}$$

A.3 Training recipes

Table 3: Training recipes. All runs use Adam, seed 42 and a CPU, with batch size 64 unless stated.

Model	Recipe
VE teacher	60 epochs, lr 10^{-3} with cosine decay, EMA 0.999
VE teacher without EMA (ablation)	40 epochs, constant lr 10^{-3}, no EMA
RF teacher (1-RF)	40 epochs, constant lr 10^{-3}, EMA 0.999
2-RF (reflow)	60 000 pairs from 100 Euler steps of the 1-RF (EMA); 10 epochs from the 1-RF weights, lr $2 \cdot 10^{-4}$, EMA 0.999
CM-VE, CM-RF, CM-RF (reflow)	CD, 40 000 iterations, constant lr $4 \cdot 10^{-4}$, $N = 150$, $\mu = 0.95$, pseudo-Huber, $\sigma_{\text{d}} = 0.5$, sampling EMA 0.9999
CM-VE (CD, no-EMA teacher) (ablation)	as CM-VE, distilled from the VE teacher without EMA
CM-VE (CT v1)	20 000 iterations, lr $4 \cdot 10^{-4}$ with cosine decay, $N(k)$ from 2 to 151, $\mu_0 = 0.95$ ($\mu(k)$ up to 0.9993), pseudo-Huber, sampled with the target network
CM-VE (CT v2)	40 000 iterations, constant lr $4 \cdot 10^{-4}$, $N(k)$ from 2 to 151, $\mu_0 = 0.9$ ($\mu(k)$ up to 0.9986), pseudo-Huber, sampling EMA 0.9995
CM-VE (iCT v1, stage 1)	20 000 iterations, lr $4 \cdot 10^{-4}$ with cosine decay, no EMA target, N from 11 to 161 in 5 stages, lognormal noise levels, weighted pseudo-Huber, sampling EMA 0.999
CM-VE (iCT v1)	stage 1 plus 32 289 iterations from its weights, batch size 128, lr $2 \cdot 10^{-4}$ with cosine decay, N from 161 to 641 in 3 stages, otherwise as stage 1
CM-VE (iCT v2)	40 000 iterations, constant lr $4 \cdot 10^{-4}$, N from 11 to 161 in 5 stages, otherwise as stage 1 except for a sampling EMA of 0.9995

A.4 Consistency training curricula and loss

CT. At iteration k of K, the number of discretization points and the EMA rate of the target network are

$$N(k) = \left\lceil \sqrt{\frac{k}{K}\left((s_1+1)^2 - s_0^2\right) + s_0^2} - 1 \right\rceil + 1, \qquad \mu(k) = \exp\left(\frac{s_0 \ln \mu_0}{N(k)}\right), \tag{8}$$

with $s_0 = 2$, $s_1 = 150$ and $\mu_0 = 0.95$ (v1) or 0.9 (v2), so that training starts with a coarse grid and refines it up to $N(K) = 151$. The grid is the Karras grid in σ with $N(k)$ points.

iCT. The number of points is $N(k) = \min\left(s_0\, 2^{\lfloor k/K' \rfloor},\ s_1\right) + 1$ with $K' = \lfloor K/(\log_2 \lfloor s_1/s_0 \rfloor + 1) \rfloor$, so that N doubles at the start of each of the equal stages: $(s_0, s_1) = (10, 160)$ in stage 1 and iCT v2 and $(160, 640)$ in the second phase of iCT v1. On the same Karras grid $\sigma_1 < \cdots < \sigma_N$, the pair n is drawn with probability

$$p_n \propto \operatorname{erf}\left(\frac{\ln \sigma_{n+1} - P_{\text{mean}}}{\sqrt{2}\, P_{\text{std}}}\right) - \operatorname{erf}\left(\frac{\ln \sigma_n - P_{\text{mean}}}{\sqrt{2}\, P_{\text{std}}}\right), \qquad P_{\text{mean}} = -1.1,\ P_{\text{std}} = 2, \tag{9}$$

and the loss is $\frac{1}{\sigma_{n+1} - \sigma_n}\, d\left(f_\theta(x_{t_{n+1}}, t_{n+1}),\ \operatorname{sg}[f_\theta(x_{t_n}, t_n)]\right)$, with sg the stop-gradient (there is no separate target network) and d the pseudo-Huber distance. As in CT, both points share the same noise z.

A.5 Sampling time

Figure 3 shows that the total sampling time grows roughly linearly with the number of images. The speed-ups of the main text hold for batches of ten or more images; for a single image, fixed overheads make them smaller.

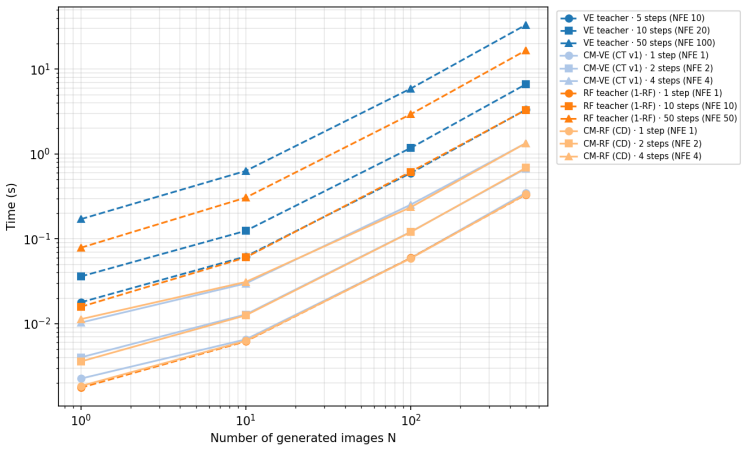


Figure 3: Sampling time versus number of generated samples N (log-log scale). Dashed lines are teachers and solid lines consistency models. Only one model per kind is drawn: models with the same network and number of steps share the same times.

A.6 Re-noising time for RF

Figure 4 shows the sweep used to choose $\tau_{\max}$ for multistep sampling with CM-RF, generated with `system/tests/cm_tau_sweep.py`. Up to $\tau_{\max} = 0.3$ the samples are almost the same as with one step, and from 0.7 on the model regenerates the digit from heavy noise and many digits change or break.

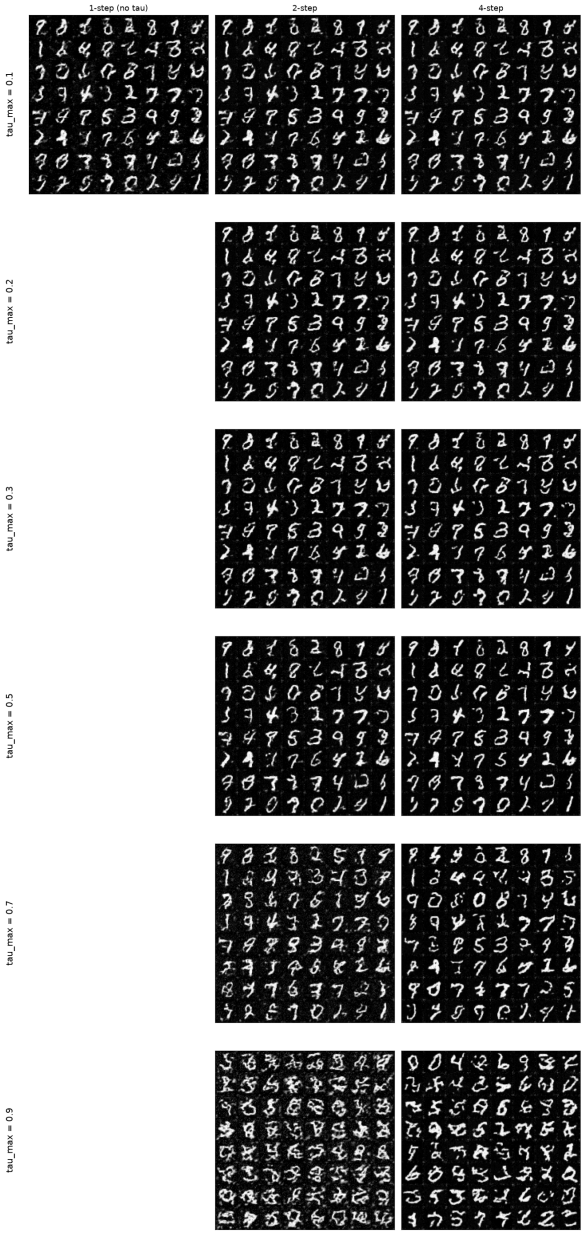


Figure 4: CM-RF samples for different values of $\tau_{\max}$. Each row is a value of $\tau_{\max}$ (0.1, 0.2, 0.3, 0.5, 0.7 and 0.9) and each column a number of steps (2 and 4); the panel at the top left is the one-step result, which does not depend on $\tau_{\max}$. Every panel shows the same 64 initial noises.

A.7 Complete results

Table 4 gives every configuration, including those left out of Table 2. Table 5 and Figure 5 give the share of each digit, which is the detail behind the entropy column.

Table 4: Complete results. Same columns as Table 2. Each consistency model is listed after the teacher it was distilled from (CM-VE from the VE teacher, CM-RF from the 1-RF, CM-RF (reflow) from the 2-RF); CT and iCT have no teacher. The times of models with the same network and steps are the same by construction (Section 4.6).

Model	Steps	NFE	FD-MNIST	Recogn. (%)	Entropy (%)	ms/img	Speed-up
Real MNIST (test)	–	–	45.1	97.6	100	–	–
Pure noise	–	–	2947.5	61.0	23	–	–
VE teacher	5	10	902.2	53.2	85	6.64	×10.0
VE teacher	10	20	120.4	79.9	98	13.26	×5.0
VE teacher	50	100	125.7	80.8	98	66.43	×1.0
CM-VE (CD)	1	1	418.3	65.4	97	0.70	×95.4
CM-VE (CD)	2	2	345.8	68.0	97	1.35	×49.3
CM-VE (CD)	4	4	279.4	73.5	98	2.68	×24.8
CM-VE (CT v1)	1	1	1922.4	38.5	50	0.70	×95.4
CM-VE (CT v1)	2	2	1481.6	36.6	77	1.35	×49.3
CM-VE (CT v1)	4	4	910.6	50.9	87	2.68	×24.8
CM-VE (CT v2)	1	1	453.5	65.5	98	0.70	×95.4
CM-VE (CT v2)	2	2	352.5	69.6	98	1.35	×49.3
CM-VE (CT v2)	4	4	267.7	74.0	99	2.68	×24.8
CM-VE (iCT v1, stage 1)	1	1	691.6	57.2	94	0.70	×95.4
CM-VE (iCT v1, stage 1)	2	2	427.4	64.2	95	1.35	×49.3
CM-VE (iCT v1, stage 1)	4	4	248.0	72.6	97	2.68	×24.8
CM-VE (iCT v1)	1	1	423.8	66.1	97	0.70	×95.4
CM-VE (iCT v1)	2	2	269.8	71.1	97	1.35	×49.3
CM-VE (iCT v1)	4	4	174.4	78.1	98	2.68	×24.8
CM-VE (iCT v2)	1	1	417.6	65.4	98	0.70	×95.4
CM-VE (iCT v2)	2	2	269.7	72.3	98	1.35	×49.3
CM-VE (iCT v2)	4	4	163.3	76.8	99	2.68	×24.8
RF teacher (1-RF)	1	1	3791.3	0.0	14	0.66	×50.1
RF teacher (1-RF)	10	10	151.9	78.4	99	6.61	×5.0
RF teacher (1-RF)	50	50	78.5	83.0	100	33.26	×1.0
RF teacher (2-RF)	1	1	533.6	61.4	98	0.66	×50.1
RF teacher (2-RF)	10	10	186.4	74.4	99	6.61	×5.0
RF teacher (2-RF)	50	50	160.0	76.2	99	33.26	×1.0
CM-RF (CD)	1	1	417.0	66.4	98	0.67	×49.6
CM-RF (CD)	2	2	341.6	73.4	99	1.37	×24.2
CM-RF (CD)	4	4	263.6	74.3	99	2.68	×12.4
CM-RF (reflow)	1	1	394.3	66.6	99	0.67	×49.6
CM-RF (reflow)	2	2	521.1	68.7	99	1.37	×24.2
CM-RF (reflow)	4	4	536.8	70.1	99	2.68	×12.4

Table 5: Percentage of samples assigned to each digit by the reference classifier (a perfectly balanced generator gives 10% for each digit).

Model	Steps	0	1	2	3	4	5	6	7	8	9
Real MNIST (test)	–	9	12	11	10	11	9	9	10	9	10
Pure noise	–	0	0	82	2	0	0	0	0	15	0
VE teacher	5	4	0	26	17	8	10	3	4	22	6
VE teacher	10	11	4	16	13	8	11	9	8	12	9
VE teacher	50	12	4	16	12	8	10	10	8	11	9
CM-VE (CD)	1	9	3	16	14	8	12	8	9	13	8
CM-VE (CD)	2	10	3	15	14	7	14	8	10	12	8
CM-VE (CD)	4	12	5	14	12	8	12	8	10	10	10
CM-VE (CT v1)	1	0	0	52	5	2	2	0	0	36	3
CM-VE (CT v1)	2	2	1	23	6	13	7	2	1	39	5
CM-VE (CT v1)	4	5	1	18	9	9	10	7	3	31	8
CM-VE (CT v2)	1	11	4	15	15	7	12	8	9	11	8
CM-VE (CT v2)	2	11	4	15	14	7	12	9	9	11	9
CM-VE (CT v2)	4	12	5	13	13	7	12	9	9	9	10
CM-VE (iCT v1, stage 1)	1	7	2	20	14	9	14	7	8	14	6
CM-VE (iCT v1, stage 1)	2	8	2	19	14	8	13	7	8	15	6
CM-VE (iCT v1, stage 1)	4	10	3	16	14	7	13	8	8	12	9
CM-VE (iCT v1)	1	9	4	16	15	8	13	8	9	11	8
CM-VE (iCT v1)	2	10	4	15	15	8	12	8	9	11	8
CM-VE (iCT v1)	4	11	4	15	14	7	12	9	8	10	9
CM-VE (iCT v2)	1	8	5	15	15	8	13	7	10	12	8
CM-VE (iCT v2)	2	9	5	15	15	8	13	7	10	12	8
CM-VE (iCT v2)	4	11	5	14	13	8	12	8	9	10	10
RF teacher (1-RF)	1	1	1	0	92	0	7	0	0	0	0
RF teacher (1-RF)	10	6	8	10	12	8	11	8	11	11	14
RF teacher (1-RF)	50	8	10	10	11	9	10	9	11	10	12
RF teacher (2-RF)	1	7	8	11	16	8	13	6	12	10	11
RF teacher (2-RF)	10	8	9	11	12	9	11	7	12	10	12
RF teacher (2-RF)	50	8	9	10	12	9	11	8	12	9	12
CM-RF (CD)	1	6	8	11	15	7	12	7	12	11	11
CM-RF (CD)	2	7	10	10	13	8	11	7	11	9	13
CM-RF (CD)	4	7	10	11	13	8	11	7	12	9	13
CM-RF (reflow)	1	6	9	11	12	8	12	6	13	11	12
CM-RF (reflow)	2	7	11	9	13	8	11	7	13	9	13
CM-RF (reflow)	4	7	12	9	12	8	11	7	13	7	14

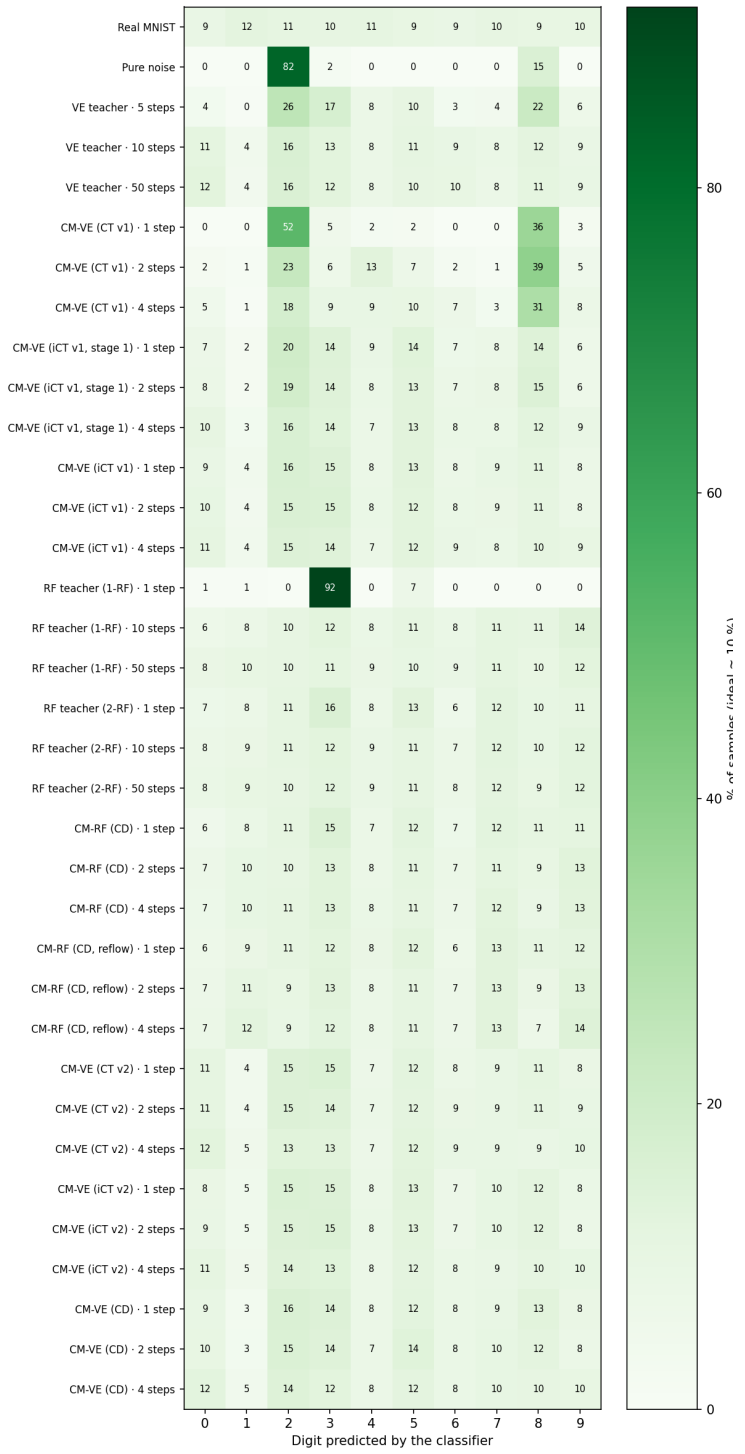


Figure 5: Distribution of the digits predicted by the reference classifier (graphical version of Table 5).

A.8 Sampling variability

Each model was trained once, but the same trained model evaluated on another batch of initial noise gives slightly different metrics. Table 6 repeats the evaluation of the consistency models with five noise seeds (other than the one of the main tables), generated with `system/evaluation/seed_variance.py`. The standard deviations are 2 to 13 for FD-MNIST, at most 1.7 points for the recognizable fraction and at most 1.2 for the entropy, so we do not interpret FD differences below about 25. The values in the main tables use another noise seed (42), which is somewhat favourable: most of them, including the one-step ones, are below these means.

Table 6: Mean and standard deviation over five noise seeds (2000 samples each) for the consistency models.

Model	Steps	FD-MNIST	Recogn. (%)	Entropy (%)
CM-VE (CD)	1	432.1 ± 7.1	65.0 ± 1.2	97.2 ± 0.3
CM-VE (CD)	2	355.6 ± 7.5	66.8 ± 0.5	97.4 ± 0.3
CM-VE (CD)	4	275.6 ± 5.0	71.9 ± 0.4	98.5 ± 0.2
CM-VE (CT v1)	1	1926.9 ± 8.1	38.8 ± 0.6	49.8 ± 1.2
CM-VE (CT v1)	2	1480.8 ± 9.1	37.3 ± 1.0	76.2 ± 1.0
CM-VE (CT v1)	4	907.3 ± 5.5	52.8 ± 1.4	87.7 ± 0.4
CM-VE (CT v2)	1	470.0 ± 7.4	63.5 ± 1.1	97.4 ± 0.4
CM-VE (CT v2)	2	365.1 ± 7.1	67.6 ± 0.7	97.6 ± 0.3
CM-VE (CT v2)	4	271.2 ± 2.4	73.6 ± 0.8	98.4 ± 0.3
CM-VE (iCT v1, stage 1)	1	705.6 ± 7.8	56.2 ± 1.3	94.1 ± 0.7
CM-VE (iCT v1, stage 1)	2	435.3 ± 5.8	64.1 ± 0.7	94.6 ± 0.5
CM-VE (iCT v1, stage 1)	4	260.3 ± 9.5	71.4 ± 0.8	96.7 ± 0.5
CM-VE (iCT v1)	1	438.1 ± 10.7	65.0 ± 1.7	96.9 ± 0.2
CM-VE (iCT v1)	2	283.3 ± 9.9	69.5 ± 0.9	97.3 ± 0.1
CM-VE (iCT v1)	4	177.0 ± 4.8	76.6 ± 1.3	98.0 ± 0.3
CM-VE (iCT v2)	1	429.3 ± 12.0	65.0 ± 1.6	97.5 ± 0.2
CM-VE (iCT v2)	2	274.8 ± 8.0	69.8 ± 1.1	97.8 ± 0.4
CM-VE (iCT v2)	4	164.2 ± 1.6	76.8 ± 0.8	98.6 ± 0.3
CM-RF (CD)	1	435.4 ± 10.7	65.7 ± 0.5	98.2 ± 0.3
CM-RF (CD)	2	353.7 ± 8.1	70.7 ± 1.6	99.0 ± 0.3
CM-RF (CD)	4	272.7 ± 5.0	73.6 ± 0.7	99.0 ± 0.2
CM-RF (reflow)	1	410.6 ± 9.4	66.4 ± 0.8	98.5 ± 0.2
CM-RF (reflow)	2	533.7 ± 10.3	68.1 ± 1.3	99.0 ± 0.1
CM-RF (reflow)	4	549.9 ± 13.2	69.9 ± 0.9	98.7 ± 0.2

A.9 Visual samples with ten initial noises

Figures 6 and 7 extend Figure 2 to all the configurations and ten initial noises. Within a figure, each row starts from the same noise, so the columns can be compared image by image.

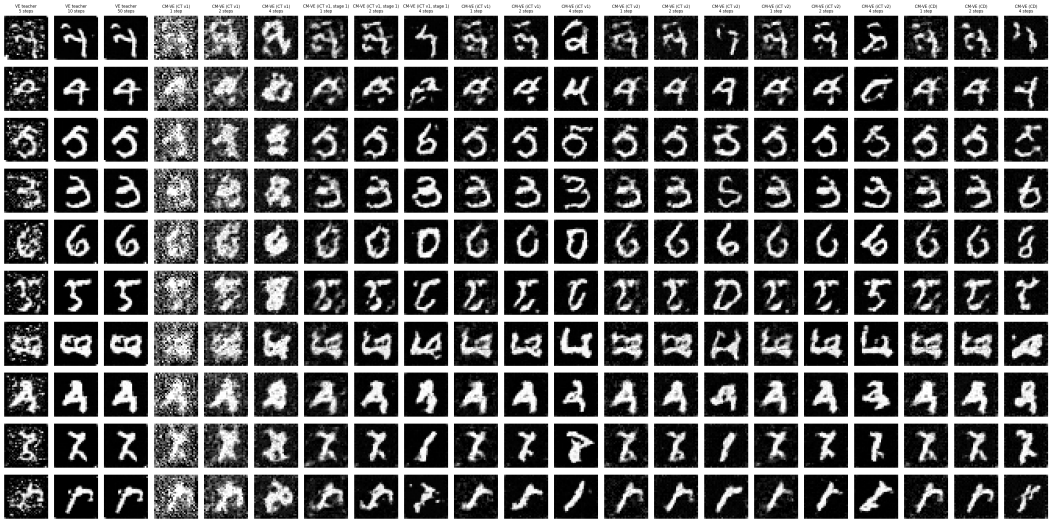


Figure 6: Samples from the VE family. Each column is a configuration and each row starts from the same initial noise.

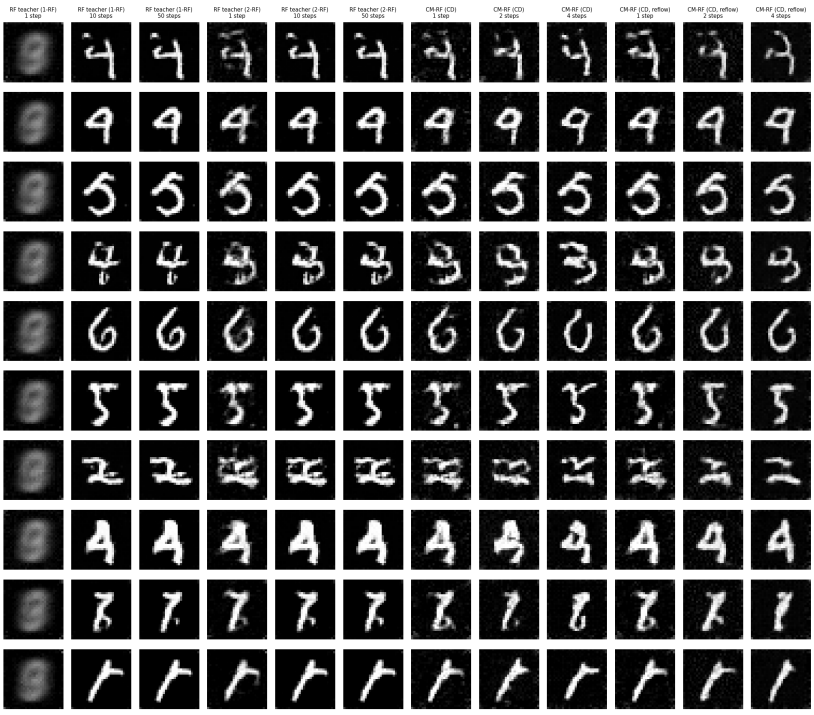


Figure 7: Samples from the RF family. Each column is a configuration and each row starts from the same initial noise.

A.10 EMA and raw teacher weights for CM-RF

Table 7 compares CM-RF distilled from the EMA and from the raw weights of the 1-RF, both evaluated on the same noise, generated with `system/evaluation/cm_rf_teacher_weights.py`. The samples look similar (Figure 8), but the EMA weights give a lower FD with every number of steps. The difference is 16, 29 and 12, so only the two-step one exceeds the threshold of 25 we use (Section 5). The entropy is also about one point higher with EMA. Each model was trained once, so this is not a general result. The one-step EMA rows are the same as those of CM-RF in Table 6; the multistep ones differ slightly (for example 354.4 against 353.7 with two steps) because the noise added in multistep sampling comes from a different random state in each script.

Table 7: CM-RF distilled from the EMA or the raw weights of the 1-RF. Mean and standard deviation over five noise seeds (2000 samples each).

Teacher weights	Steps	FD-MNIST	Recogn. (%)	Entropy (%)
EMA	1	435.4 ± 10.7	65.7 ± 0.5	98.2 ± 0.3
EMA	2	354.4 ± 6.1	71.3 ± 0.8	99.0 ± 0.3
EMA	4	276.0 ± 4.8	73.3 ± 1.3	99.0 ± 0.2
Raw	1	451.7 ± 13.0	64.9 ± 1.7	96.9 ± 0.4
Raw	2	382.9 ± 10.2	69.2 ± 1.4	97.7 ± 0.3
Raw	4	287.5 ± 6.1	72.1 ± 0.7	97.9 ± 0.1

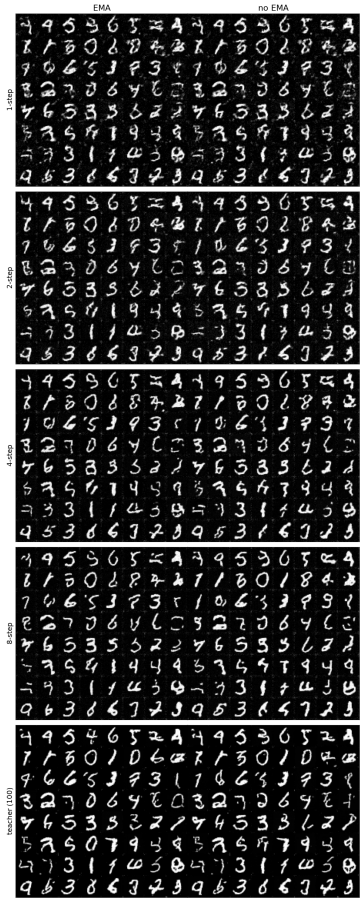


Figure 8: CM-RF distilled from the EMA weights of the 1-RF (left) and from its raw weights (right), with 1, 2, 4 and 8 steps and the 100-step teacher (bottom). All panels start from the same noises.

A.11 VE teacher without EMA

Our first VE teacher was trained for 40 epochs with a constant learning rate and without EMA; the one used in the main text was trained for 60 epochs with cosine decay and EMA (Section 4.2). Two things changed at once, so we cannot say which one matters. Table 8 gives the quality of both teachers and of a CM-VE distilled from each one with the same recipe. The EMA teacher is clearly better (FD 126 against 206 with 50 steps, 39% lower) and gives more ones (4% against 2% of the samples; real MNIST has 12%). The CM-VE distilled from it improves by 36, 47 and 61 FD with one, two and four steps (differences of the five-seed means), more than the sampling noise (standard deviations of 5 to 8), but in proportion much less than the teacher: 8 to 18% lower. With one and two steps the samples of the two models look similar (Figure 9); with four steps, in these five noises, the model from the EMA teacher breaks or changes more digits (the 3 and the 6) even though its FD is lower, which five noises are not enough to interpret. The rest of the report only uses the EMA teacher, which is the one the reported CM-VE was distilled from.

Table 8: VE teacher with and without EMA, and the CM-VE distilled from each one. Teachers: one noise seed, 50 steps. Consistency models: mean and standard deviation over five noise seeds (2000 samples each).

Model	FD-MNIST	Recogn. (%)	Entropy (%)
VE teacher (50 steps)	125.7	80.8	97.8
CM-VE (CD) (1 step)	432.1 ± 7.1	65.0 ± 1.2	97.2 ± 0.3
CM-VE (CD) (2 steps)	355.6 ± 7.5	66.8 ± 0.5	97.4 ± 0.3
CM-VE (CD) (4 steps)	275.6 ± 5.0	71.9 ± 0.4	98.5 ± 0.2
VE teacher (no EMA) (50 steps)	205.7	79.4	95.2
CM-VE (CD, no-EMA teacher) (1 step)	467.8 ± 5.3	64.5 ± 1.0	94.9 ± 0.4
CM-VE (CD, no-EMA teacher) (2 steps)	402.5 ± 6.6	67.0 ± 1.0	95.2 ± 0.2
CM-VE (CD, no-EMA teacher) (4 steps)	336.8 ± 7.3	72.1 ± 1.8	95.5 ± 0.5

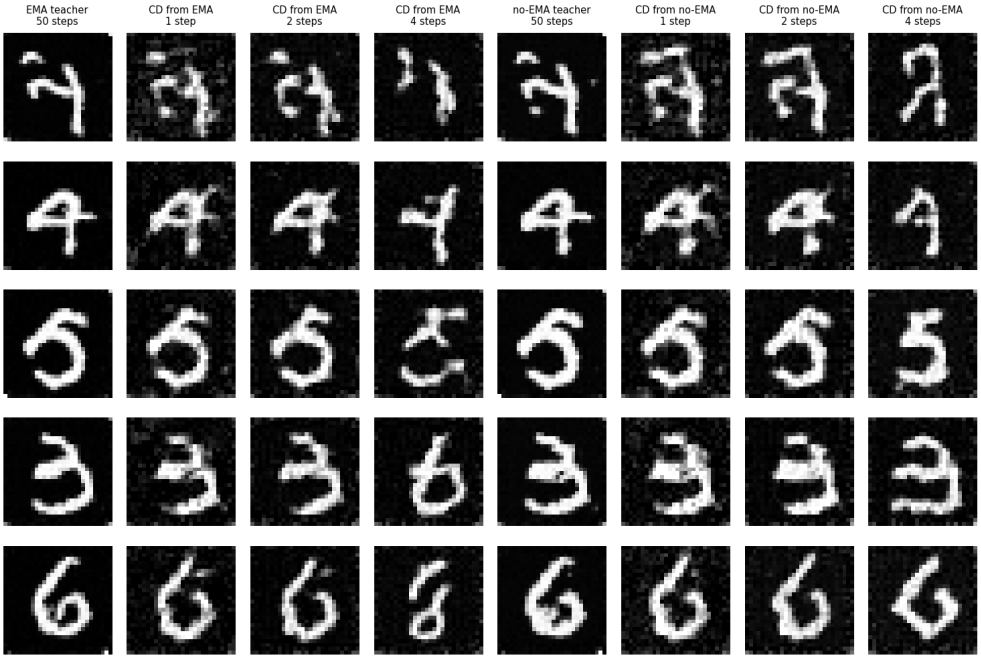


Figure 9: VE teacher and CM-VE distilled from it, with EMA (left) and without EMA (right). Each row starts from the same initial noise.

Author Contributions

Roles follow the CRediT taxonomy. **Ibon de Mingo Arroyo**: Conceptualization, Methodology, Software (code structure, VE diffusion teacher, consistency distillation, consistency training and improved consistency training), Investigation, Writing – original draft. **Eloísa Denia García**: Methodology, Software (Rectified Flow teacher, reflow, consistency distillation from Rectified Flow, CT v2 and iCT v2 runs), Investigation, Visualization (RF ablations), Writing – original draft. **Matin Esmaeilzadeh**: Methodology (evaluation protocol), Software (reference classifier, metrics and comparison pipeline), Formal analysis, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration regarding the use of AI tools

During the preparation of this manuscript, the authors used Claude (Anthropic; Claude Opus 5.5 and Claude Sonnet 5.5) for AI-assisted coding and language editing. The use of the tool was limited to helping write and debug parts of the code (training scripts and the evaluation pipeline), checking the numbers and claims of the report against the code and the results, and editing the wording of the report and the notebook.